



GROUP TECHNICAL SOLUTIONS (GTS)

PETRONAS CHEMICALS LDPE SDN BHD

Plant Step Test Procedure

September 2020

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	Name:	Signature:	Role:	Date:
Prepared by:	M Aminuddin Ishak		Engineer Process Control	20/09/2020
Reviewed and Approved by:	Andriny Rusman		Principal Engineer Process Control	20/09/2020

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1.0 INTRODUCTION

PETRONAS Chemicals LDPE Sdn Bhd has embarked on a project to revamp its Advanced Process Control application. This involves reviewing design of the advanced controller, inferential and the grade transition programme. The new APC controller are expected to be commissioning between by Q3 of 2020.

2.0 OBJECTIVE

The objectives of plant step test are:

- To perturb the plant APC MVs/DVs and observed response of CVs within normal operating envelope
- To generate process data to be used for APC model identification

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3.0 ABBREVIATIONS

i.	APC	Advanced Process Control
ii.	AUT	Automatic (Controller Mode)
iii.	CV	Controlled Variable
iv.	DV	Disturbance Variable
v.	FDS	Functional Design Specifications
vi.	MAN	Manual (Controller Mode)
vii.	MFI	Melt Flow Index
viii.	MV	Manipulated Variable
ix.	PCLDPE	PETRONAS Chemicals LDPE Sdn Bhd
x.	PIMS	Plant Information Management System
xi.	TTSS	Time to Steady State, T

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4.0 METHODOLOGY

4.1 Plant Step Test

Plant step test is carried out before APC model identification stage. The step test is conducted to obtain data for model identification.

4.1.1 Preparation

The lists of template / document below should be made available before plant step test activities:

- i. APC step test operation engagement pack (i.e. MV/DV & CV list, presentation slides)
- ii. Basic APC Controller Functional Design and Strategies (FDS)
- iii. APC Matrix (Finalized after FDS review workshop)
- iv. Trip and alarm settings
- v. Pre Test Procedure
- vi. Risk Assessment
- vii. Permit to Work
- viii. Step test logbook
- ix. Template for data collection
- x. Previous related technical report for references

4.1.2 Pre-requisite Condition

Before conducting the plant step test on each MVs/DVs, the states of the base layer control or any existing regulatory control are assessed and rectified if possible before/during plant step test. This includes controller scheme modification, controller automation, controller tuning and functional analyzer with good quality measurement (i.e. MFI). Any relevant instrumentation issues, hardware / process constraint / limitation must be addressed before and during step test activities.

Process data historian (PIMS: PI OSI server) must be in good working condition for trending monitoring and data collection. PIMS administrator is required to be on standby to attend to any technical difficulties faced by PIMS during plant step test/step test.

Management approval must be obtained before commencement of step test. Engagement sessions with the operation should be conducted prior to or during the step test.

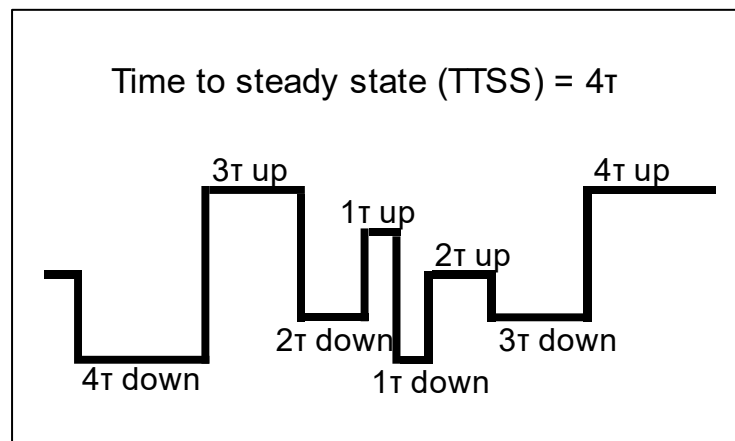
The plant should be running at sufficient steady state (normal operating condition with 20% tolerance) with no other auxiliary unit shut down, unit conducting performance test runs, major maintenance or any other

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conflicting activities, scheduled during the step test period. The plan step test shall be carried out within the normal operating envelope of the process and no major upset is expected.

4.1.3 Step Test Execution

Upon discussion with APC engineer, panel operators will be requested to make one step up and one step down on the controller's output (MAN) or controller's set point (AUTO). Subjected to process conditions, move size should be significant to overcome process noise and large enough to monitor response of related CVs but kept within normal operating limit to avoid plant upsets. Before implementation of any steps, panel operator will be consulted with regards to the step size to be made. The most suitable step size for step test will be jointly determined between plant operators and APC engineers based on the pretest data.



Operator may choose to make the move up / down based on plant conditions. A step change / move is made one at a time if plant condition permits. A move is sustained until related CVs (e.g. T, P, and Quality) settle to a new steady state (Time to Steady State - TTSS = 4τ). The move is then repeated again for 3τ , 2τ , 1τ , 1τ , 2τ , 3τ and finally 4τ . All step changes must return back to the original settings or at any settings preferred by operator before the next test (next MV). APC engineer on duty will advise panel on the duration of each step test.

If other MVs are required to be moved to normalize the process or to bring the CV to specification, operator must be allowed to do so. However, it is advisable to execute the MV moves at different time and with different moves pattern to prevent correlated moves. These moves can be treated as step test moves if uncorrelated. If panel operator needs to make such changes that intervene with the step test, APC engineer should be informed accordingly as he/she will have to decide how long to hold his/her intended moves and when to resume step test.

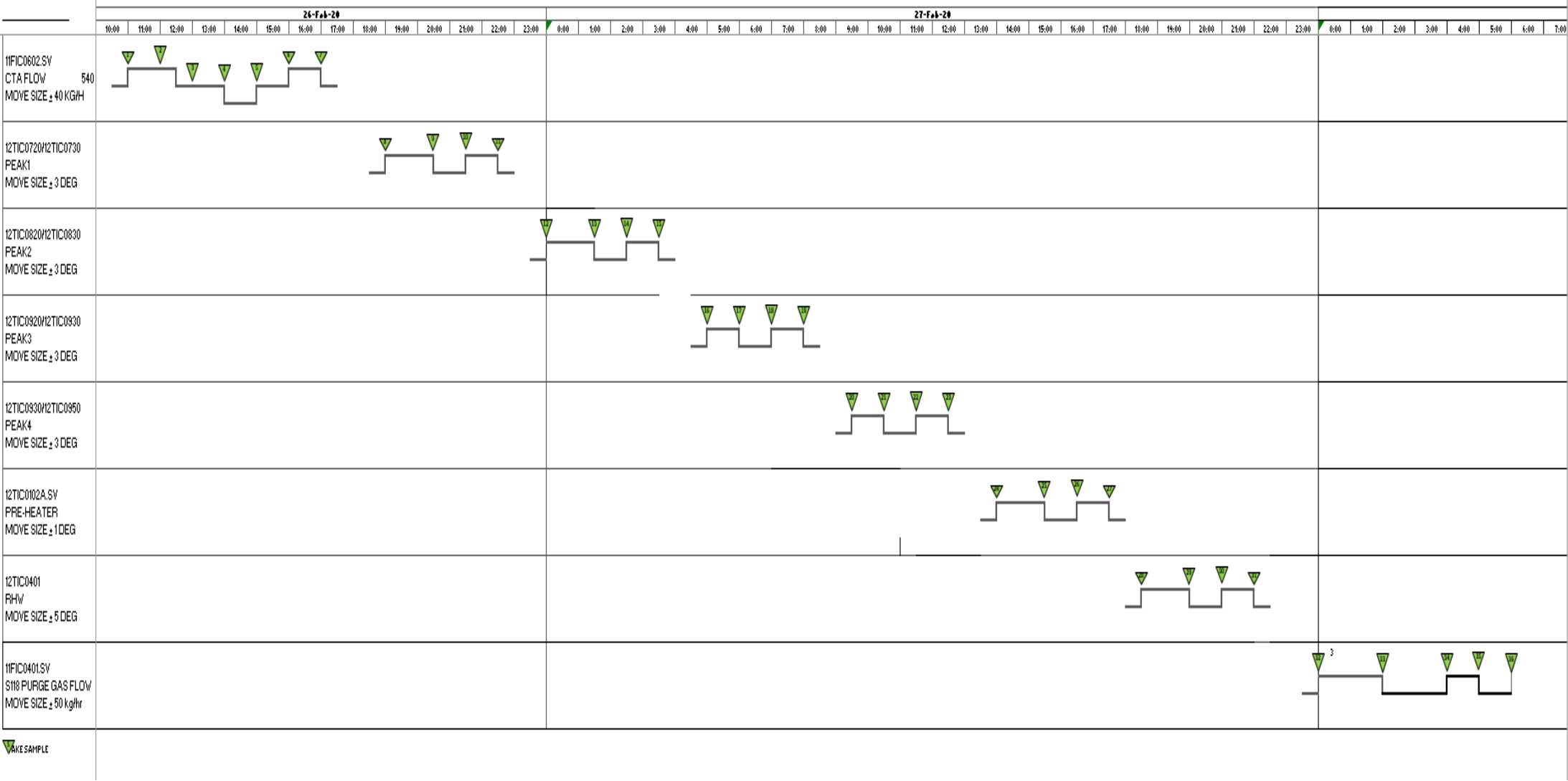
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Process trending on the possible CVs related to the MV move is monitored and recorded by the APC Engineer. In the event of unfavorable condition (as advised by the panel operator), the test for a particular MVs can be deferred.

Below is the example of proposed step test plan. However, it may be adjusted later depending on plant status and condition.



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5.0 PLANT STEPTEST PLAN

5.1 Plant Step Test Duration

PCLDPE Plant step test for C150Y product grade is planned to commence from 8th to 16th March (8 days). The plant step test would be carried out during over 24 hours. The Plant test will be carried out with two (2) 12 hrs shift as below:

Shift 1: 8:30 am to 8:30 pm,

Shift 2: 8:30 pm to 8:30 am

5.2 Plant Step Test MV/DV List

The proposed minimum step size below was based on historical data of changes made by operator and by previous step test conducted at 2008.

Total Number of MV/DV = 8

No. of MVs = 6

No. of DVs = 2

MV	Step Size by Grade											
	N103X	Base SV	N109X	Base SV	N125Y	Base SV	C150Y	Base SV	M022X	Base SV	M050X High Peak	Base SV
CTA (11FIC0602)	± 10 kg/h	250 kg/h	± 15 kg/h	365 kg/h	± 30 kg/h	440 kg/h	± 40 kg/h	540 kg/h	± 40 kg/h	850 kg/h	± 50 kg/h	980 kg/h
Peak 1 (12TIC0720/12 TIC0730)	± 3 deg	285 deg	± 3 deg	285 deg	± 3 deg	296 deg	± 3 deg	296 deg	± 3 deg	296 deg	± 3 deg	296 deg
Peak 2 (12TIC0820/12 TIC0830)	± 3 deg	285 deg	± 3 deg	290 deg	± 3 deg	296 deg	± 3 deg	296 deg	± 3 deg	296 deg	± 3 deg	296 deg
Peak 3 (12TIC0920/12 TIC0930)	± 3 deg	285 deg	± 3 deg	290 deg	± 3 deg	296 deg	± 3 deg	296 deg	± 3 deg	296 deg	± 3 deg	296 deg
Peak 4 (12TIC0930/12 TIC0950)	± 3 deg	285 deg	± 3 deg	290 deg	± 3 deg	296 deg	± 3 deg	296 deg	± 3 deg	296 deg	± 3 deg	274 deg
RHW Temp (12TIC0401)	± 5 deg	160 deg	± 5 deg	160 deg	± 5 deg	160 deg	± 5 deg	160 deg	± 3 deg	160 deg	± 3 deg	160 deg
12TIC0102A Preheater	± 2 deg	167deg	± 2 deg	167 deg	± 2 deg	167 deg	± 2 deg	167 deg	± 2 deg	167 deg	± 2 deg	167 deg
11FIC0401 (Purge)					± 100 deg		± 100 deg					



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6.0 APC MATRICES

6.1 PCLDPE APC Matrix

CV		MFI	GLOSS	PRODSPE X	APTEMP 1	APTEMP 2	APTEMP 3	APTEMP 4
		Melt Flow Index	Gloss	Production Rate	Peak Temp 1	Peak Temp 2	Peak Temp 3	Peak Temp 4
MV								
APCTA	Propylene Flow	MV (+ve)	DV (+ve)	DV (+ve)				
APTEMP1SV	Filtered P Temperature 1 PV	DV (+ve)	MV (-ve)	MV (+ve)	MV (+ve)			
APTEMP2SV	Filtered P Temperature 2 PV	DV (+ve)	MV (-ve)	MV (+ve)		MV (+ve)		
APTEMP3SV	Filtered P Temperature 3 PV	DV (+ve)	MV (-ve)	MV (+ve)			MV (+ve)	
APTEMP4SV	Filtered P Temperature 4 PV	DV (+ve)	MV (-ve)	MV (+ve)				MV (+ve)
11FIC0401	Purge gas		MV (-ve)					
12PIC0103.SV	Reactor Pressure	DV (-ve)	DV (+ve)					
12TIC0401.PV	RHW Temperature (Filtered PV)	DV (+ve)	DV (+ve)	DV (-ve)				
12TIC0102A.PV	Reactor Start Temperature	DV (+ve)	DV (+ve)	DV (-ve)				
APTEMP1	Peak Temperature 1	DV (+ve)						
APTEMP2	Peak Temperature 2	DV (+ve)						
APTEMP3	Peak Temperature 3	DV (+ve)						



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APTEMP4	Peak Temperature 4	DV (+ve)						
RXDIFTEMPTOT	Total temperature difference	DV(-ve)						
CTARATIO	CTAFLOW/PRODSPEX PV	DV (+ve)						
PRODSPEXPV	Production Rate (Filtered PV)	DV (-ve)						



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7.0 PLANT STEP TEST LOG

Plant Step test and step test log should be made available to log plant test activities during the entire duration. This to ensure seamless communication between process control engineers while conducting the test to avoid unintended incidences. Completed step test and step test log will be attached in the step test and step test report.

7.1 Plant Step test Log

APC Engineer On Duty	Date	Tag	Activities							Remarks
			4 Tau (Mins)	Tau	Duration	Time	From	To	Step Size	
	Plant status									
	Sunday									
	Plant Status									
	Monday									



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8.0 RISK ASSESSMENT

8.1 Risk Assessment Matrix

REFER ATTACHMENT